Detection of cancer through urine fluorescence

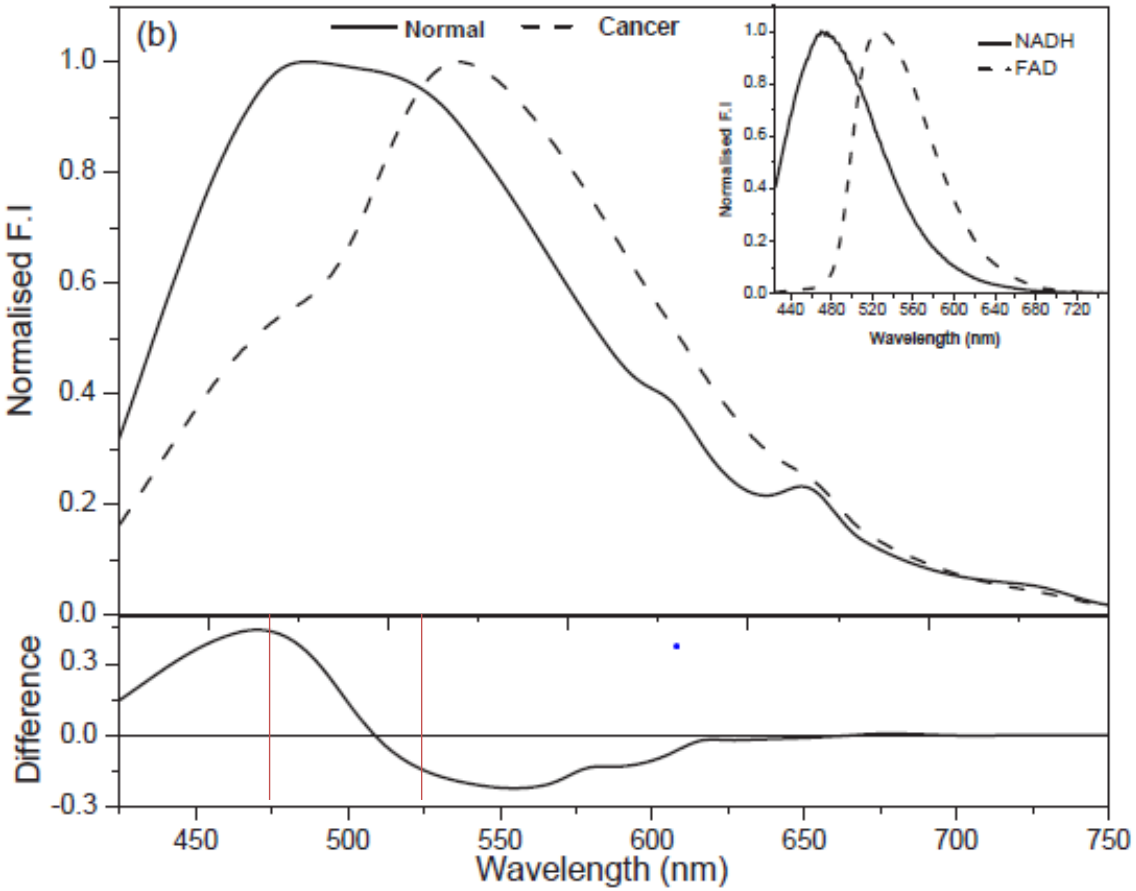
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Problem description

Many metabolites in human urine (e.g. NADH) are native fluorophores.

Fluorescent intensity values of the emission spectra at different wavelengths are significantly different in the healthy population and in patients with cancer.

Fluorescence intensity spectra are valid diagnostic parameters for screening of cancer, but spectrofluorometers are very expensive (e.g. DUETTA, Horiba. 24000€). This limits the clinical applications.



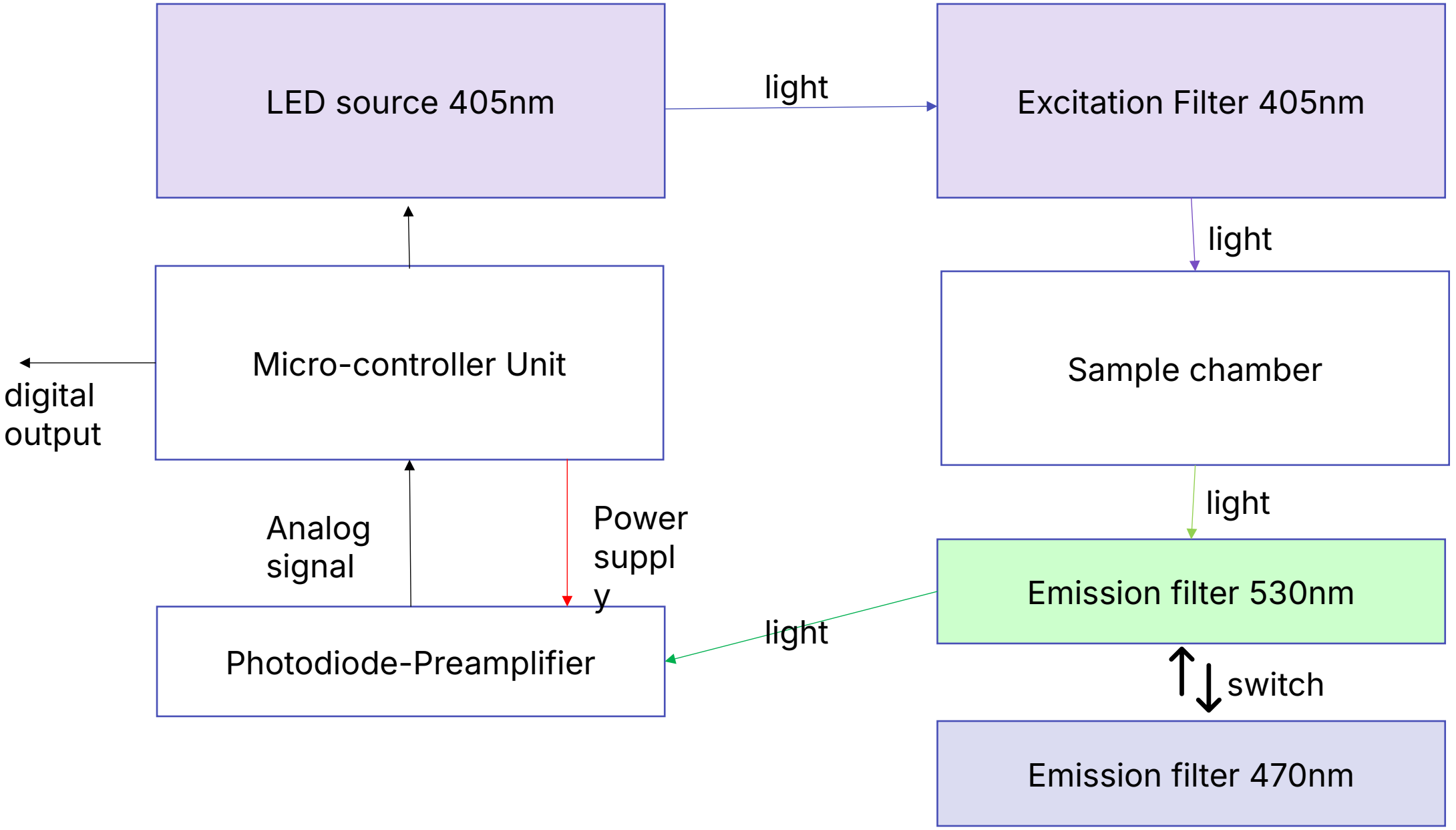
Fluorescence Intensity emission spectra when stimulated at 405nm

Instrumentation rationale

Researchers* showed that measuring emission at two different wavelengths is enough to reach high sensitivity (>90%): low-cost optoelectronic systems to detect cancer based on urine fluorescence are possible.

The sample is illuminated at 405nm using a LED and the fluorescence intensity is measured at 470nm and 530nm using a photodiode.

*Characterization and Diagnosis of Cancer by Native Fluorescence Spectroscopy of Human Urine, Rajasekaran et al. 2012


LED source 405nm
light
Excitation Filter 405nm
light
Sample chamber
light
Emission filter 530nm
switch
Emission filter 470nm
light
Photodiode-Preamplifier
Analog signal
Power supply
Micro-controller Unit
digital output

Component	Model	Important characteristics	Price
LED	LED405E, Thorlabs	Optical Output Power: 10mW Peak wavelength: 405nm FWHM=15n	17,12€
Excitation filter	FB405-10, Thorlabs	Peak = 405 ± 2nm FWHM = 10 ± 2nm	98.36 €
Emission filter 1	FB470-10, Thorlabs	Peak = 470 ± 2nm FWHM = 10 ± 2nm	92.46 €
Emission filter 2	FB530-10, Thorlabs	Peak = 530 ± 2nm FWHM = 10 ± 2nm	85.09 €
Photodiode	ODA-6W-500M, OptoDiode	Linear response range: 400-960nm	~60 €

~353€